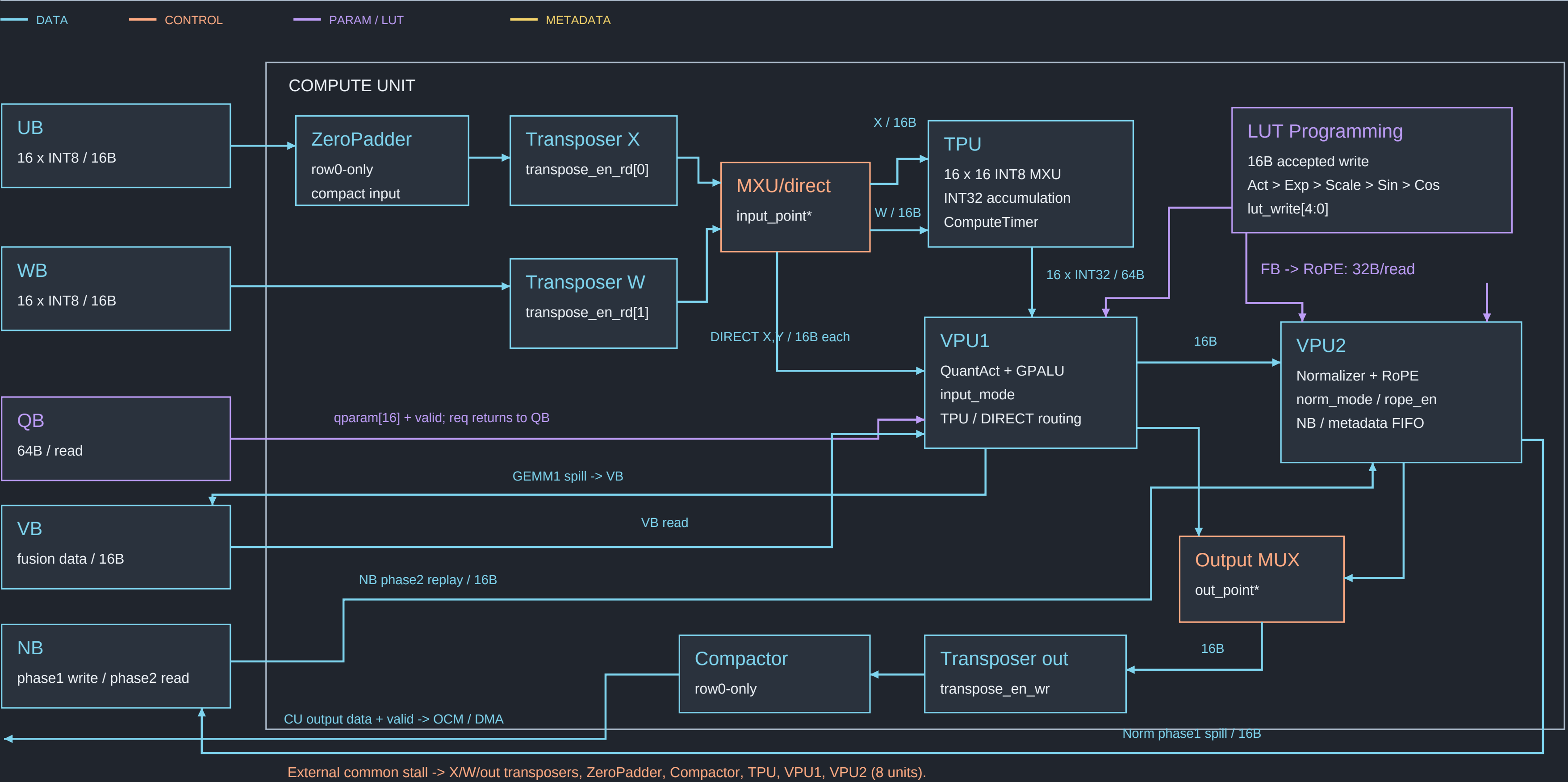


Compute Unit / block routing

Target architecture from npu(1).pdf + later frozen contracts. Selectors marked * are diagram aliases; encodings are not inferred.



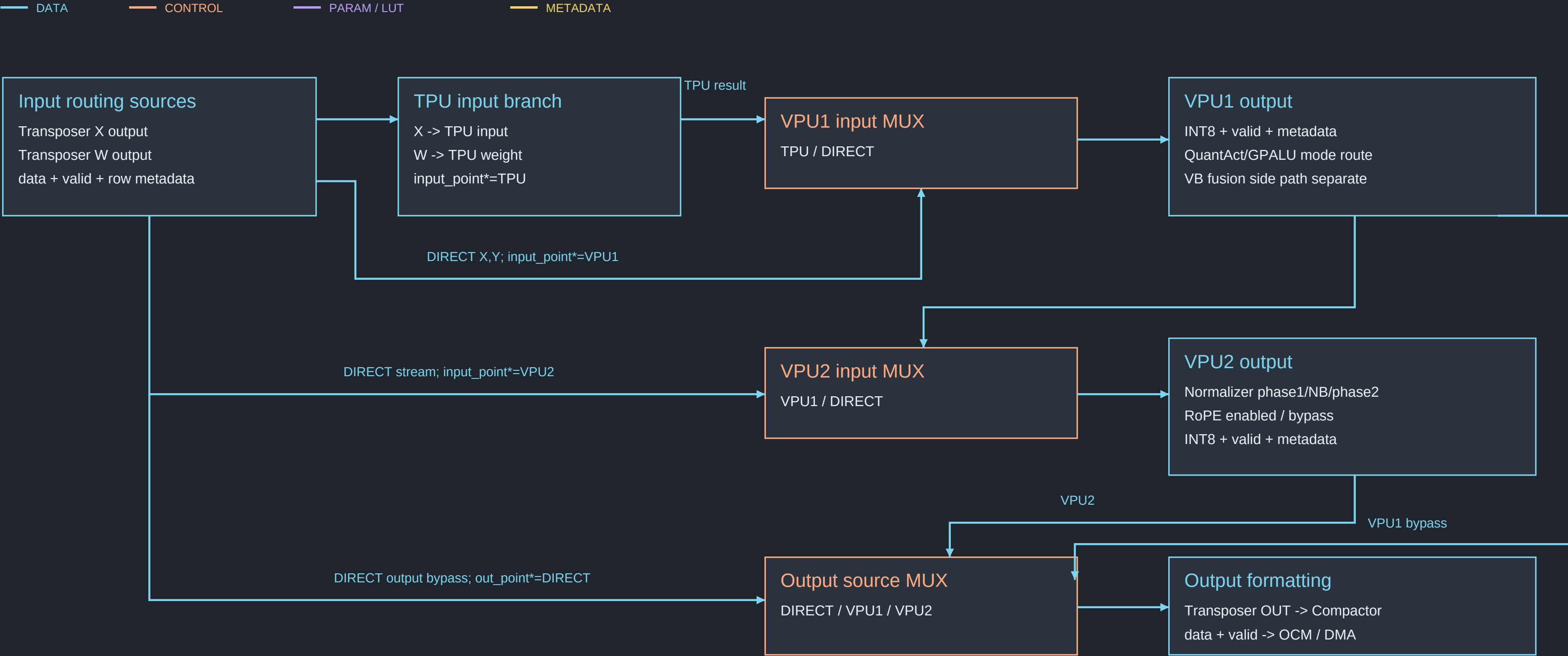
Metadata row_change_update follows the selected data path, including Norm phase1 -> FIFO -> phase2 -> RoPE.

Source: npu(1).pdf; CODEX_HANDOFF_QB64_FB32.md (2026-09-09); user routing/LUT decisions (2026-09-12).

Direct-to-VPU2 and direct-to-output bypass branches are defined on the selector sheet; omitted here to keep buses readable.

Compute Unit / bypass, metadata and common stall

Routing expansion [A/D]. * selectors are symbolic aliases; direct-stream source and numeric selection encoding require current Comp.scala.



External stall distributor -> common stall

Transposer X | Transposer W | Transposer OUT | ZeroPadder | Compactor | TPU | VPU1 | VPU2

Every MUX selects a stream bundle: data, valid, and associated metadata. No independent metadata selection is allowed.
Norm metadata FIFO bridges phase1 write and phase2 replay. Outstanding parameter/LUT responses may be captured while stalled.

Compute Unit / selection & control contract

Symbolic selection values describe routing. * = diagram alias / exact top-level RTL mapping remains to be verified.

DATA

CONTROL

PARAM / LUT

METADATA

Selector / control	Symbolic choice	Selected data / action
transpose_en_rd[0], [1]	0 / 1	X or W bypass / transpose (independent instances)
input_point*	TPU / VPU1 / VPU2	Inject the selected input stream at the chosen compute stage; exact enum TBD
input_mode	TPU	TPU INT32 -> QuantAct -> GPALU(+VB) -> INT8
input_mode	DIRECT	UB/WB INT8 -> GPALU forced ADD -> INT10 -> QuantAct -> INT8
vpu1_a_sel*, vpu1_b_sel*	derived from input_mode	TPU: QuantAct, VB/constant; DIRECT: X, Y/constant
qact_src_sel*	derived from input_mode	TPU: accumulator INT32; DIRECT: GPALU INT10
vpu1_out_sel*	derived from input_mode	TPU: GPALU result; DIRECT: QuantAct result
param_mode / matrix_param	PER_MATRIX / PER_CHANNEL	one UInt32 broadcast / one QB 64B group; DIRECT never requests QB
act_mask / fusion_second	operation / fusion phase	Per-pass activation selection; one activation LUT per lane
alu_mode / out_shift	operation / output scaling	TPU path operation as configured; DIRECT forces ADD
alu_constant_operand_enable	0 / 1	stream operand / constant operand; sign/width mapping requires current RTL
norm_mode_sel	0 BYPASS / 1 RMS / 2 LN / 3 SM	Normalizer datapath and statistics mode (saved Norm source)
layout_mode	ONLINE / DISTRIBUTED	Per-lane statistics / cross-lane reduction; vector boundary differs
rope_en	0 / 1	RoPE bypass / rotate eight feature pairs
out_point*	DIRECT / VPU1 / VPU2	Selected INT8 stream -> output transposer -> row0 compactor
transpose_en_wr	0 / 1	Output bypass / transpose
vector_compact_in/out	0 / 1	Dense bypass / row0-only expansion or selection
lut_write[4:0]	Act, Exp, Scale, Sin, Cos	[4], [3], [2], [1], [0]; skip every disabled target
stall	one common signal	Freeze accepted-beat state; retain already-issued memory responses

Routing selectors must move data, valid, and row-change metadata together. Changing input_mode requires a drained pipeline.

No numeric encoding is assigned to * selectors. Existing rs1 field names are historical labels, not a verified current register map.

Fusion VB write/read direction follows the configured pass. Width conversion and disabled-path valid gating belong at each selected boundary.